

SP ENGINEERING ACADEMY

Video Lectures / Course Materials

COMPLETE COURSE

DIGITAL SYSTEMS

DIGITAL CIRCUIT DESIGN

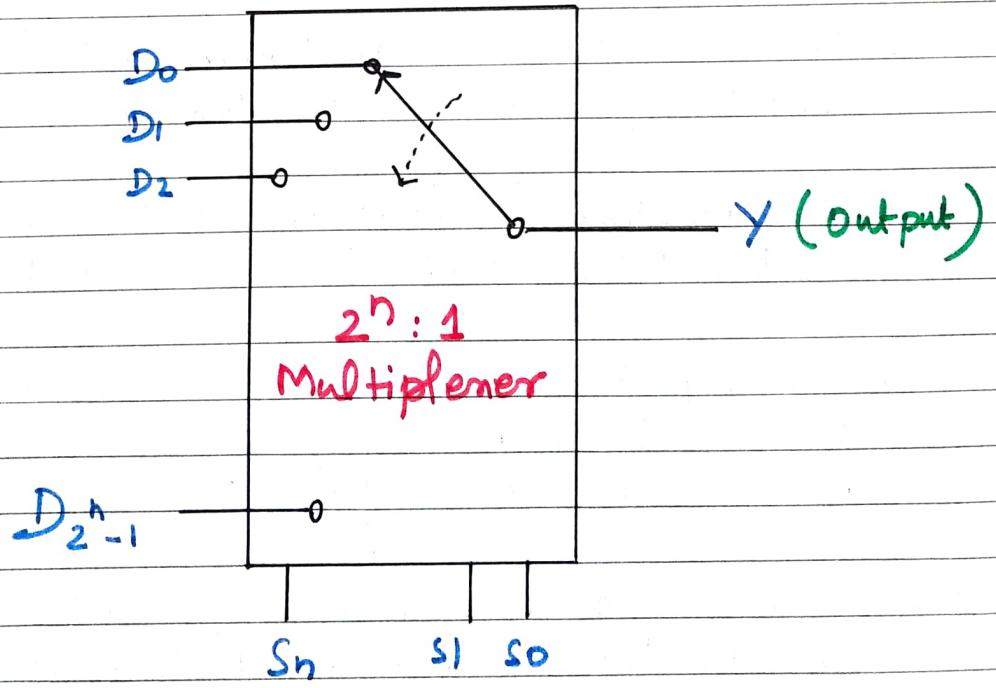
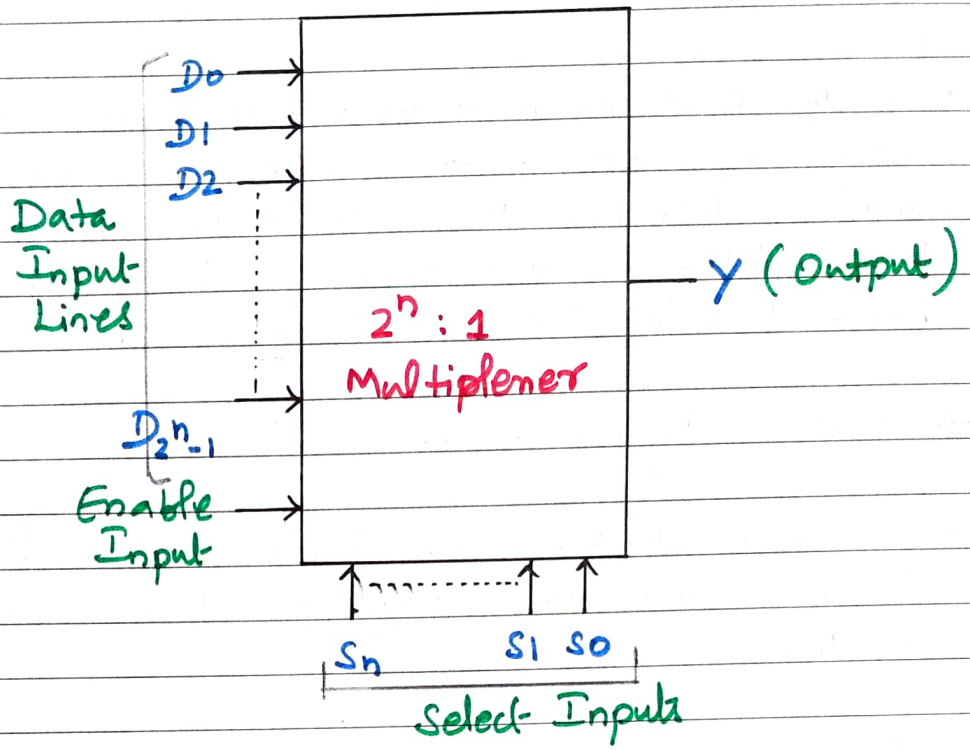
Lecture Series #11 - Multiplexers

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Multiplexers

Video Link - <https://youtu.be/kJIm7YQ08A8>

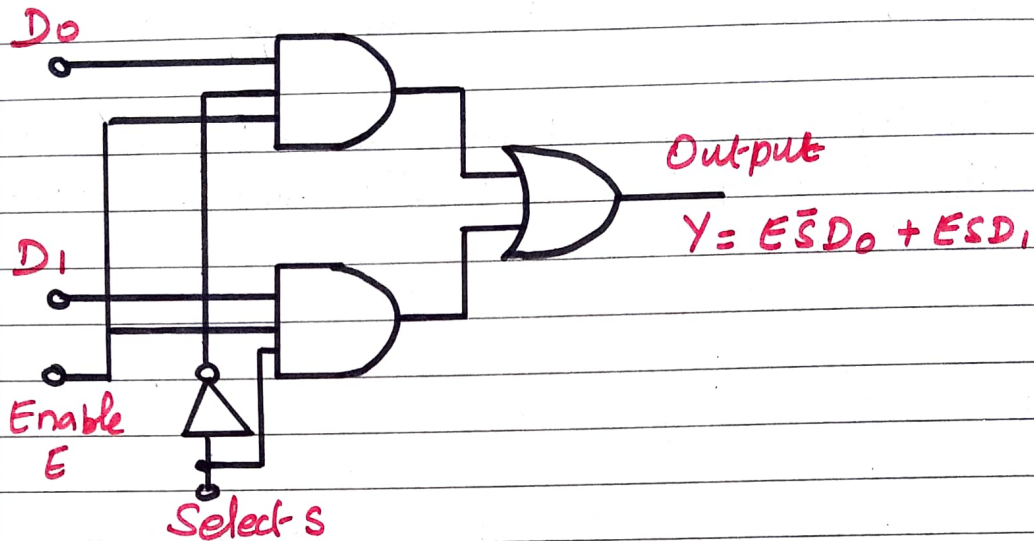
- In digital system, many times it is necessary to select single data line from several data-input lines, and the data from the selected data line should be available on the output. The digital circuit which does this task is a multiplexer.
- It is a digital switch.
- It allows digital information from several sources to be routed onto a single output line, as shown in diagram.
- The basic multiplexer has several data-input lines and a single output line.
- The selection of a particular input line is controlled by a set of selection lines.
- Since multiplexer selects one of the inputs and routes it to output, it is known as data selector.
- Normally, there are 2^n input lines and n selection lines whose bit combinations determine which input is selected.
- Therefore, multiplexer is many to one and it provides the digital equivalent of an analog selector switch.



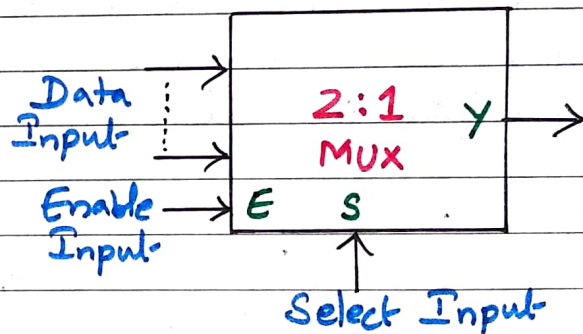
2:1 Multiplexer

Video Link - <https://youtu.be/fjDIgrFiG0M>

Diagram shows 2:1 multiplexer



E	S	Y
1	0	D ₀
1	1	D ₁
0	X	0



Enable E	Select S	D ₁	D ₀	Output Y
1	0	X	0	0
1	0	X	1	1
1	1	0	X	0
1	1	1	X	1
0	X	X	X	0

$\bar{E} S D_0$

$E S D_1$

When $E=0$, output Y is 0, irrespective of any Input Condition

When $E=1$, the Circuit works as follows

When $S=0$, the inverted S , that is 1 gets applied as second input to first AND gate. Since S is applied directly as the input to second AND gate, its output

goes zero irrespective of first input. Since the second input of first AND gate is 1, its output is equal to its first input, that is D_0 . Hence $Y = D_0$.

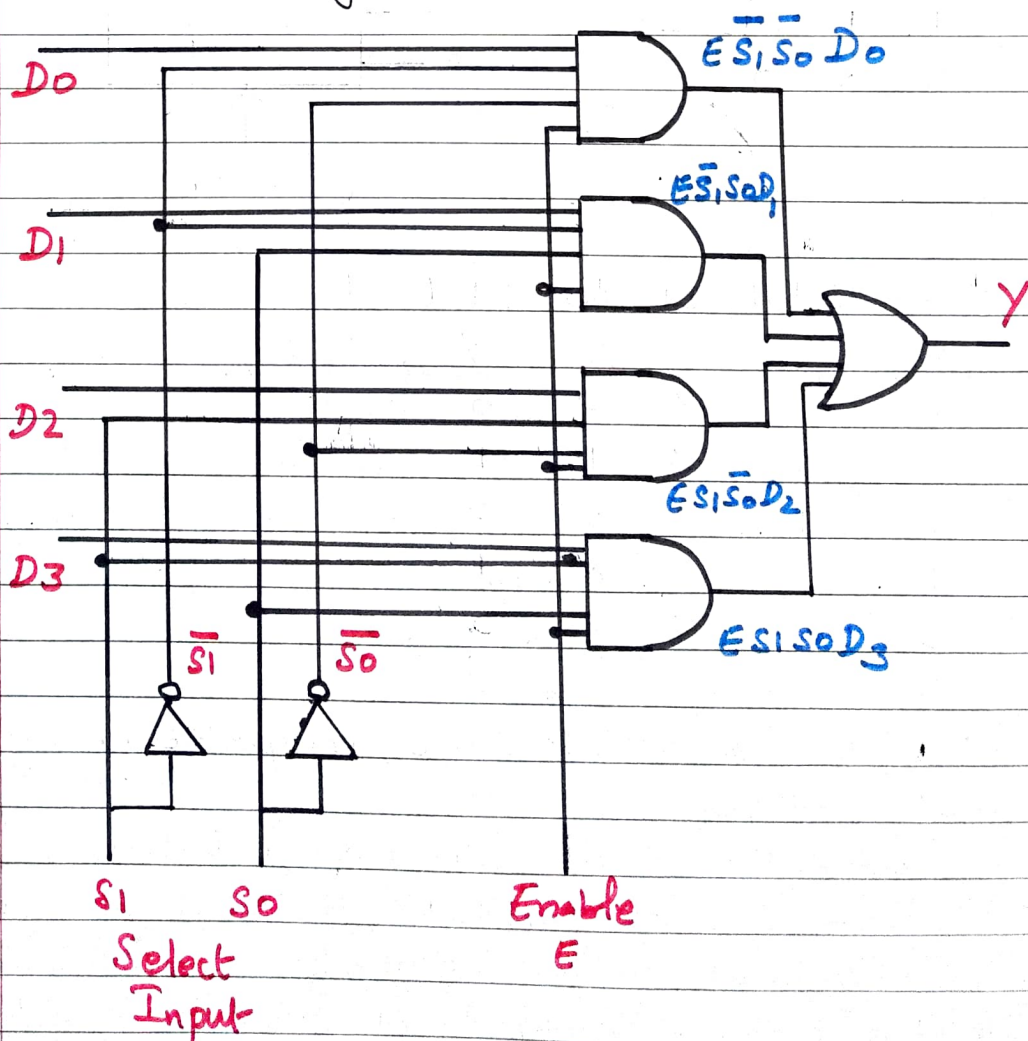
Exactly opposite is the case when $s = 1$.

In this case, second AND gate output is equal to its first input D_1 and first AND gate output is Zero. Hence $Y = D_1$.

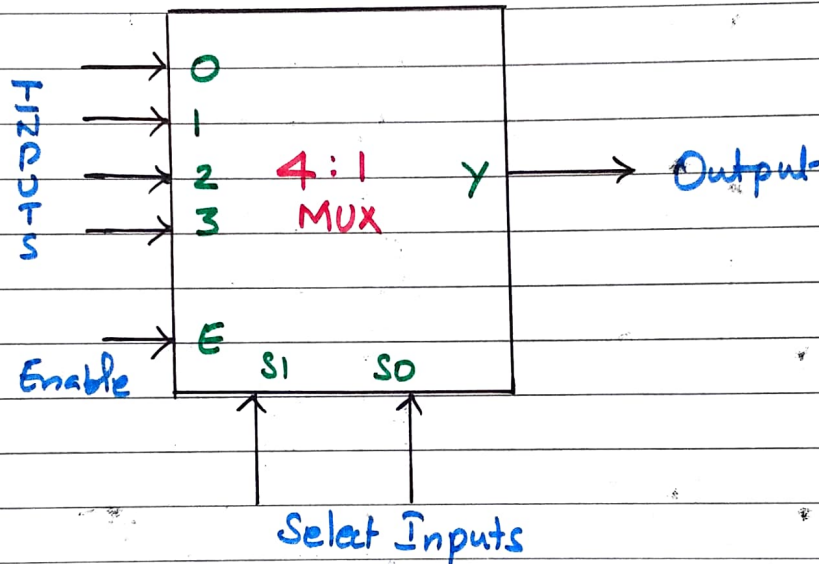
4 : 1 Multiplexer

Video Link - <https://youtu.be/Bho7gJXjBDI>

Diagram Shows the 4:1 multiplexer. Each of the four lines D_0 to D_3 is applied to one input of an AND gate. Selection lines are decoded to select a particular AND gate.



Enable	Select Inputs		Output
E	S ₁	S ₀	Y _i
1	0	0	D ₀
1	0	1	D ₁
1	1	0	D ₂
1	1	1	D ₃
X	X	X	0



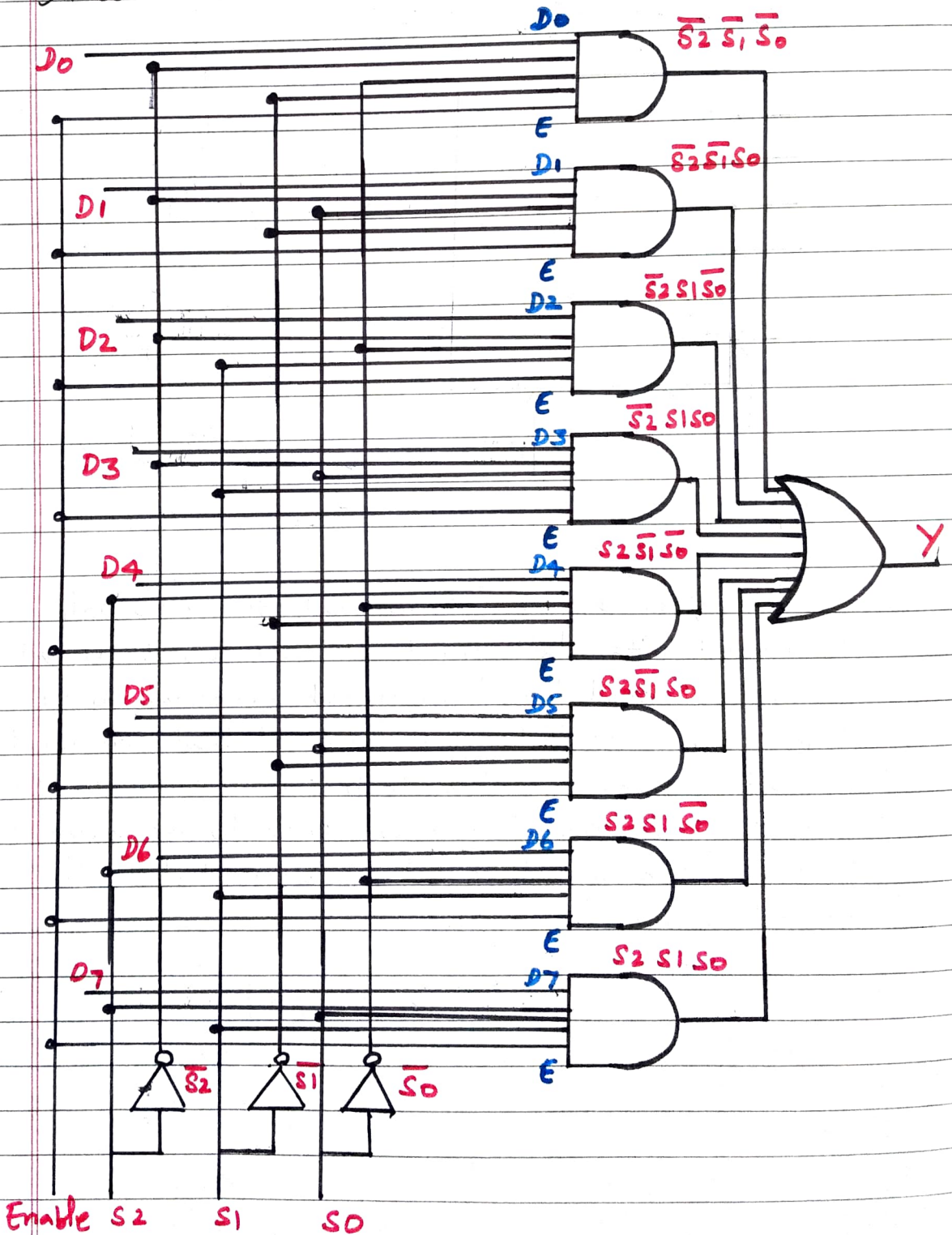
For example, when $S_1, S_0 = 01$, the AND gate associated with data input D_1 has two of its inputs equal to 1 and the third input connected to D_1 . The other three AND gates have at least one input equal to 0, which makes their outputs equal to 0. The OR gate output is now equal to the value of D_1 , thus we can say data bit D_1 is routed to the output when $S_1, S_0 = 01$.

8:1 Multiplexer

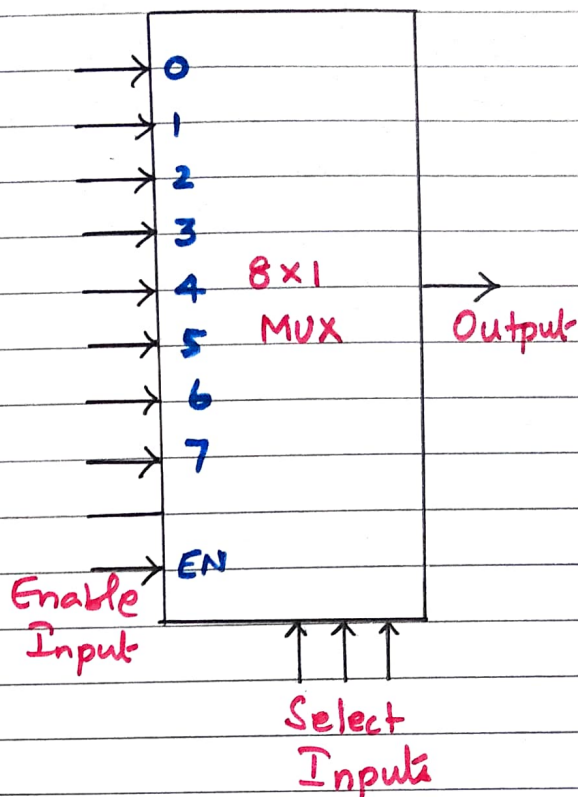
Figure shows 8:1 multiplexer

Video Link - <https://youtu.be/1EbPHcB1LhM>

There are eight input lines one output line and three select lines. As shown in table, the selection of a particular input line is controlled by three selection lines



Select Inputs			Output
S ₂	S ₁	S ₀	Y
0	0	0	D ₀
0	0	1	D ₁
0	1	0	D ₂
0	1	1	D ₃
1	0	0	D ₄
1	0	1	D ₅
1	1	0	D ₆
1	1	1	D ₇



Quadruple 2 to 1 Multiplexer Video Link - <https://youtu.be/lxArXd43DvY>

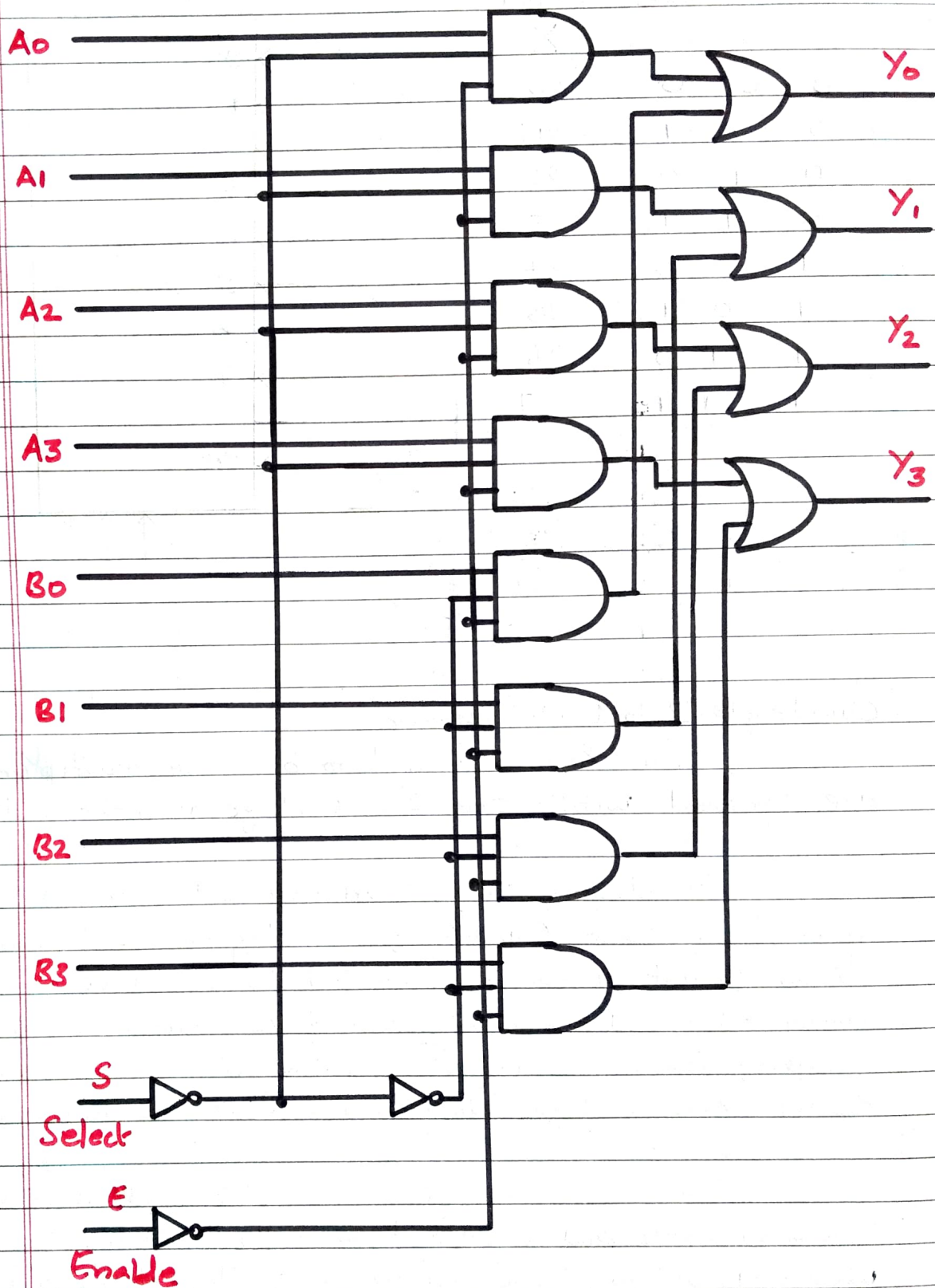
In some cases, two or more multiplexers are enclosed within one IC package as shown in diagram

The diagram shows quadruple 2 to 1 line multiplexer, each capable of selecting one of two input lines. Output Y_1 can be selected to be equal to either A_1 or B_1 . Similarly output Y_2 may have the value of A_2 or B_2 , and so on. The selection line S selects one of two lines in all four multiplexers.

The Control input E enables the multiplexers in the 0 state and disables them in the 1 state.

When $E=1$, outputs have all 0's, regardless of the values of S

E	S	Output Y
1	x	All 0's
0	0	Select A
0	1	Select B

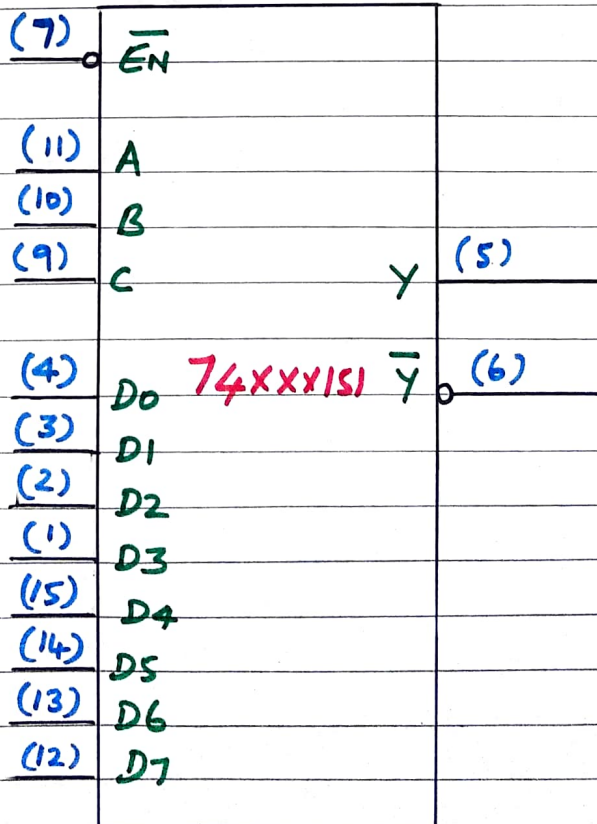


74151 Multiplexer

Video Link - <https://youtu.be/IBgnByGa0AM>

The 74xx151 is a 8 to 1 multiplexer. It has Eight inputs. It provides two outputs, one is active high, the other is active low.

Input Select			Enable	Outputs	
C	B	A	$\overline{EN}$	Y	$\overline{Y}$
x	x	x	1	0	1
0	0	0	0	D ₀	$\overline{D_0}$
0	0	1	0	D ₁	$\overline{D_1}$
0	1	0	0	D ₂	$\overline{D_2}$
0	1	1	0	D ₃	$\overline{D_3}$
1	0	0	0	D ₄	$\overline{D_4}$
1	0	1	0	D ₅	$\overline{D_5}$
1	1	0	0	D ₆	$\overline{D_6}$
1	1	1	0	D ₇	$\overline{D_7}$



74xx153 Dual 4 to 1 Multiplexer

The 74xx153 is a dual 4 to 1 multiplexer

Diagram shows the logic symbol for 74xx153.

It contains two identical and independent 4 to 1 multiplexers. Each multiplexer has separate enable inputs.

Inputs				Outputs	
1EN	2EN	B	A	1Y	2Y
0	0	0	0	1D ₀	2D ₀
0	0	0	1	1D ₁	2D ₁
0	0	1	0	1D ₂	2D ₂
0	0	1	1	1D ₃	2D ₃
0	1	0	0	1D ₀	0
0	1	0	1	1D ₁	0
0	1	1	0	1D ₂	0
0	1	1	1	1D ₃	0
1	0	0	0	0	2D ₀
1	0	0	1	0	2D ₁
1	0	1	0	0	2D ₂
1	0	1	1	0	2D ₃
1	1	X	X	0	0

74xx153			
(14)	A		
(2)	B		
(1)	1EN		
(6)	1D ₀	1Y	(7)
(5)	1D ₁		
(4)	1D ₂		
(3)	1D ₃	2Y	(9)
(1)	2EN		
(10)	2D ₀		
(11)	2D ₁		
(12)	2D ₂		
(13)	2D ₃		

Example 01

Design 16:1 multiplexer using 8:1 multiplexer

Video Link - <https://youtu.be/6NRhZooAAUc>

Step 1 :

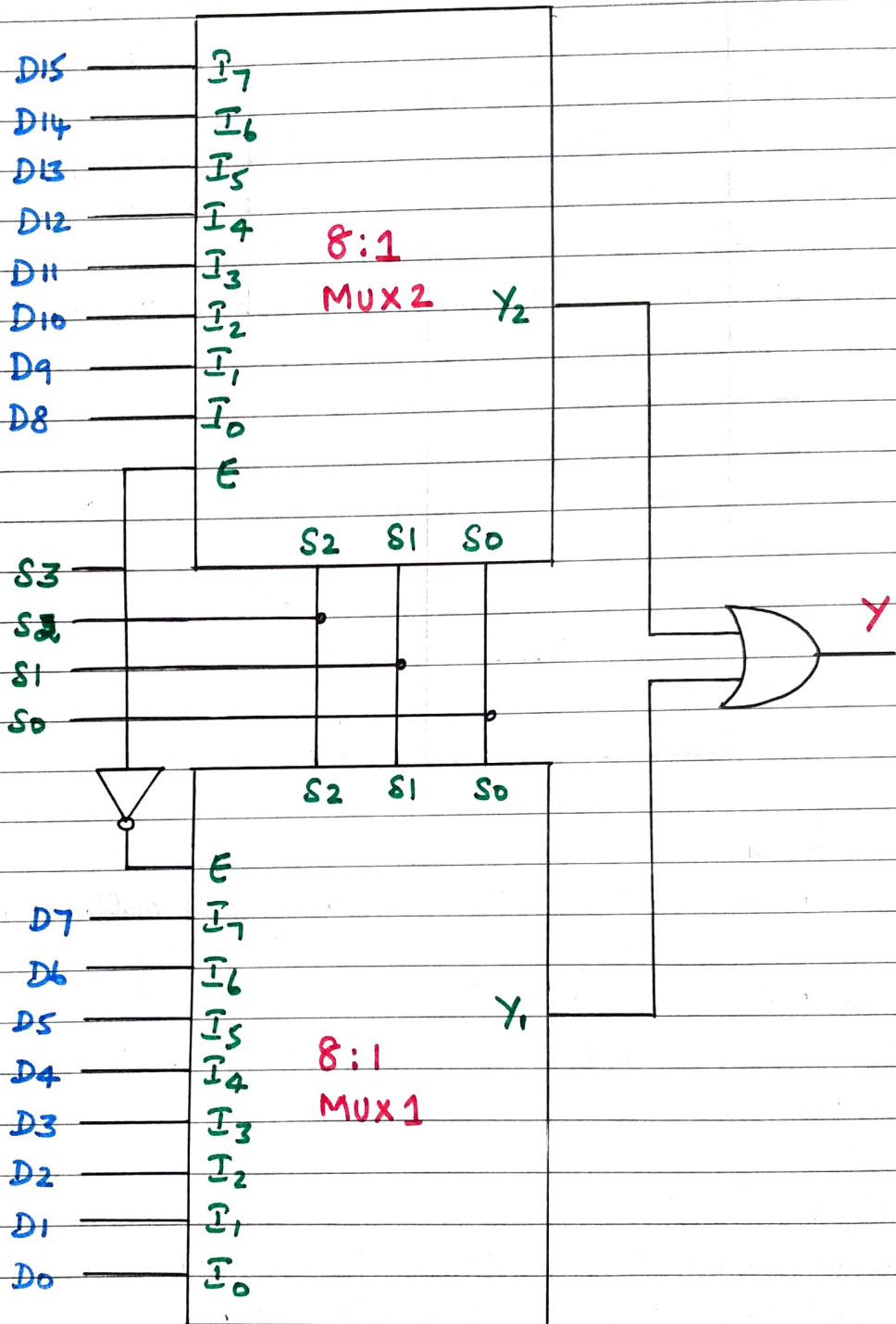
Connect the select lines (S_2 , S_1 and S_0) of two multiplexers in parallel

Step 2 :

Connect most significant select line (S_3) such that when $S_3=0$ Mux 1 is enabled and when $S_3=1$, Mux 2 is enabled

Step 3 :

Logically OR the outputs of two multiplexers to obtain the final output Y.



Example 02

Design 16:1 multiplexer using 4:1 multiplexers

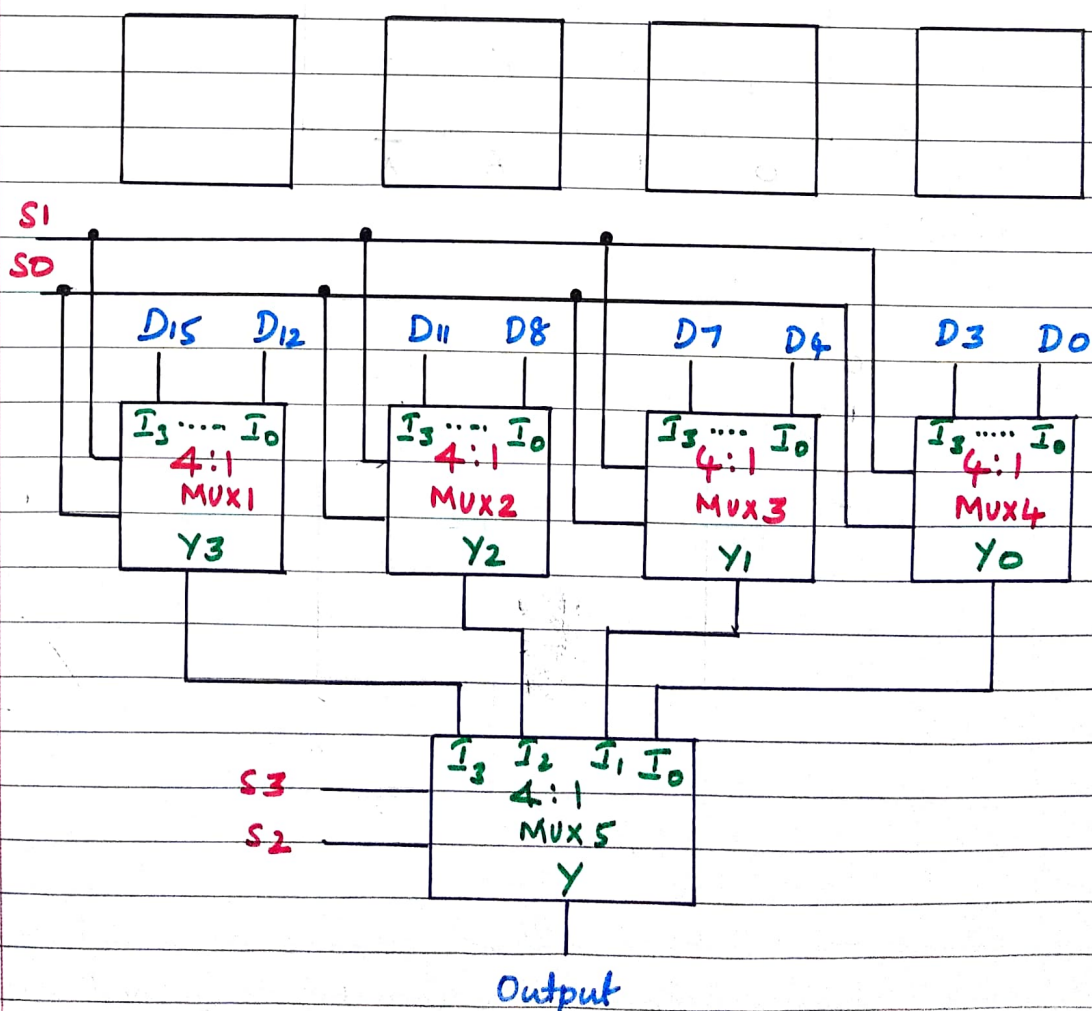
Video Link - <https://youtu.be/h291eGJQyw>

Since there are 16 inputs for the multiplexers we require four 4:1 multiplexers to satisfy input needs. The four outputs of 4:1 multiplexers are again multiplexed by 4:1 multiplexers to generate final output.

Step 1: Connect the select lines (S_1 and S_0) of four multiplexers in parallel

Step 2: Connect the most significant lines (S_3 and S_2) to the MUX 5

Step 3: Connect the outputs Y_0, Y_1, Y_2 and Y_3 of four multiplexers as data inputs for the MUX 5



Implementation of Combinational Logic using MUX

Example 01 Video Link - <https://youtu.be/la0HLEE8Z6c>

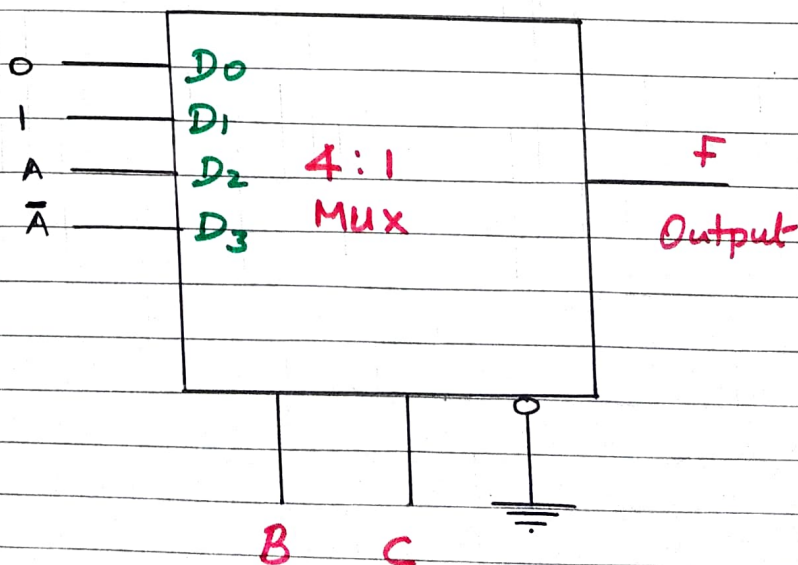
Implement the following Boolean function using 4:1 Multiplexer $F(A, B, C) = \sum m(1, 3, 5, 6)$

Step 1 - Connect least Significant Variables as a Select inputs of multiplexers. Here C to S₀ and B to S₁.

Step 2 - Derive inputs for multiplexer using implementation table

	D ₀	D ₁	D ₂	D ₃
$\bar{A}$	0	①	2	③
A	4	⑤	⑥	7
	0	1	A	$\bar{A}$

Most Significant Bit



Connect the least Significant variables as a select inputs of multiplexer. Here Connect C to S₀ and B to S₁

Example 02

Video Link - <https://youtu.be/SP5ZRshh4NU>

Implement the following Boolean function using 8:1 multiplexer

$$F(A, B, C, D) = \bar{A}B\bar{D} + ACD + \bar{B}CD + \bar{A}\bar{C}D$$

The given Boolean expression is not in Standard SOP Form. Let us first convert this in Standard SOP form

$$F(A, B, C, D) = \bar{A}B\bar{D} + ACD + \bar{B}CD + \bar{A}\bar{C}D$$

$$= \bar{A}B\bar{D}(C + \bar{C}) + ACD(B + \bar{B}) + \bar{B}CD(A + \bar{A}) + \bar{A}\bar{C}D(B + \bar{B})$$

$$= \bar{A}BC\bar{D} + \bar{A}B\bar{C}\bar{D} + ABCD + A\bar{B}CD + A\bar{B}\bar{C}D + A\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + \bar{A}\bar{B}\bar{C}D$$

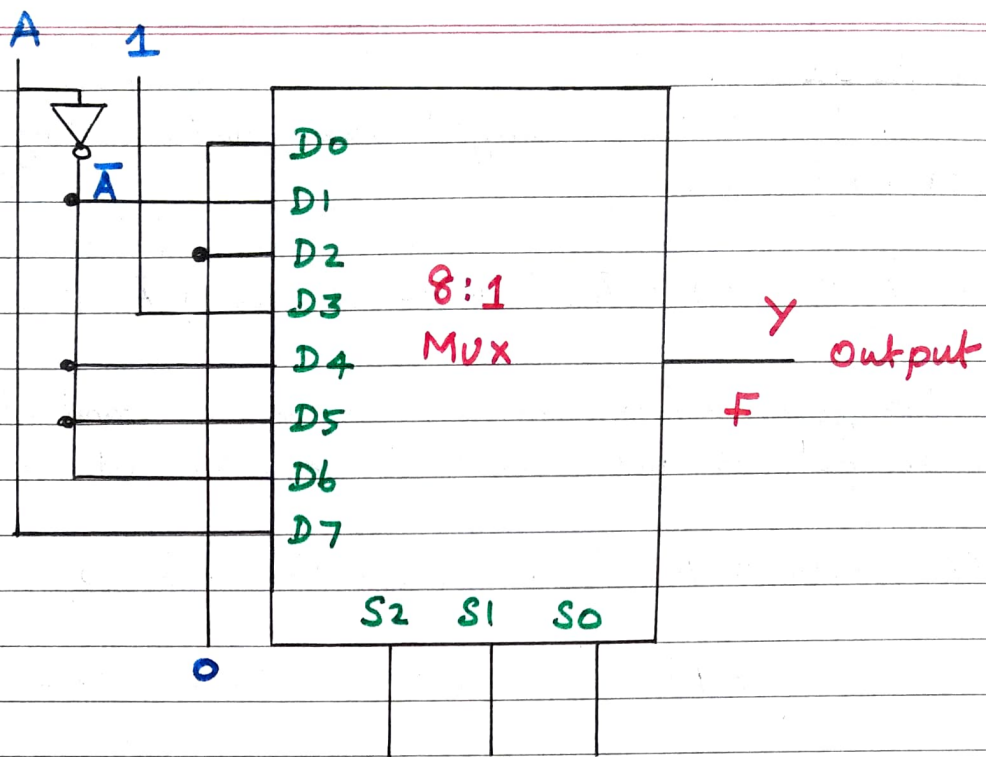
$$= \bar{A}BC\bar{D} + \bar{A}B\bar{C}\bar{D} + ABCD + A\bar{B}CD + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$= \sum m(6, 4, 15, 11, 3, 5, 1)$$

$$F(A, B, C, D) = \sum m(1, 3, 4, 5, 6, 11, 15)$$

Connect least Significant Variables as a select inputs of multiplexer. Here D to S₀, C to S₁ and B to S₂.

	D ₀	D ₁	D ₂	D ₃	D ₄	D ₅	D ₆	D ₇
$\bar{A}$	0	①	2	③	④	⑤	⑥	7
A	8	9	10	⑪	12	13	14	⑮
Most Significant Bit	0	A	0	1	A	A	A	$\bar{A}$



Applications of Multiplexers

They are used as

- Data Selector to select one out of many data inputs
- Implementation of Combinational logic circuits
- Time multiplexing systems
- Frequency multiplexing systems
- Analog to Digital and Digital to analog Converter
- Data acquisition systems

Multiplexer ICs

IC Number	Function
74150	16:1 multiplexer
74151	8:1 multiplexer
74153	Dual 4:1 multiplexer
74157	Quad 2-input multiplexer